

CASE REPORT

Open Access



Separation-delayed skin flap for diabetic foot repair: a case report and review of the literature

Aijun Guan¹ , Jiangling Yao², Yangyang Bian², Danxin Wang^{2,3}, Hong Huang², Meiting Jiang¹, Xin Cai¹, Shaowen Cheng^{2*†} and Dongyang Zeng^{1*†}

Abstract

Background Repairing diabetic foot wounds is challenging due to the high risk of necrosis associated with traditional skin flap techniques. This report introduces a novel separation-delayed flap technique and elaborates on its application in treating diabetic foot ulcers.

Case presentation A 54-year-old Chinese male patient was diagnosed with diabetic foot ulceration, exhibiting a 4 cm × 3 cm full-thickness skin defect on the lateral aspect of the left ankle, with exposed fibular cortex and extension into the tibiotalar joint cavity. Upon admission, the wound underwent debridement followed by antibiotic-impregnated bone cement placement, and 3 weeks later, a conventional peroneal fascia-based flap was created, then isolated using a rubber membrane, and sutured in situ. Following 10 days of isolation, the flap was excised and transferred to the defect site for wound repair. Postoperatively, the skin flap survived entirely, the wound healed successfully, and the function of the skin flap returned to normal.

Conclusion Separation-delayed skin flap technique increases the survival rate of skin flaps by incrementally improving blood supply, making it an effective option for repairing diabetic foot wounds.

Keywords Case report, Delayed flap, Diabetic foot, Skin defect, Wound repair

Background

The diabetes map released by the International Diabetes Federation indicates that the number of people living with diabetes in China reached 140 million in 2021 and is projected to increase to 174 million by 2045 [1]. Diabetic

foot ulcer (DFU) is one of the most prevalent clinical complications among individuals with diabetes, characterized by foot skin and tissue destruction, often accompanied by peripheral neuropathy or arterial disease [2]. Studies have demonstrated that approximately 15–25% of individuals with diabetes may develop DFU [3]. The healing of diabetic foot wounds is influenced by numerous factors, with vascular disease, neuropathy, and immune dysfunction induced by diabetes frequently leading to prolonged wound-healing difficulties. As an essential surgical intervention, flap repair has been extensively applied in wound healing. However, due to impaired blood circulation and compromised immune function in patients with diabetes, the survival rate of skin flaps and the efficacy of wound healing are often restricted [4]. In recent years, separation-delayed skin flap repair has emerged as an effective approach for treating diabetic foot wounds, owing to its advantages in enhancing

[†]Shaowen Cheng and Dongyang Zeng have contributed equally to this work.

*Correspondence:

Shaowen Cheng
chengshaowen1@126.com

Dongyang Zeng
zengdongyang@163.com

¹ School of Nursing, Hainan Medical University, Haikou 571199, China

² Key Laboratory of Emergency and Trauma of Ministry of Education, Key Laboratory of Hainan Trauma and Disaster Rescue, Department of Wound Repair, the First Affiliated Hospital of Hainan Medical University, Haikou 570102, China

³ Department of Nursing, the First Affiliated Hospital of Hainan Medical University, Haikou 570102, China



skin flap survival, improving blood supply, and promoting healing. By temporarily isolating the skin flap during the initial stage, vascular regeneration and blood supply are gradually enhanced, thereby providing optimal physiological conditions for subsequent flap transplantation. This report aims to present a successful case of delayed skin flap repair for diabetic foot wounds and to evaluate the feasibility and effectiveness of this treatment method in clinical practice.

Case presentation

A 54-year-old Chinese man was admitted for a chronic, nonhealing skin defect on the left lateral malleolus with associated purulent discharge lasting more than 4 weeks. The patient had a history of type 2 diabetes mellitus for more than 10 years, managed with long-term oral metformin and acarbose, but with poor glycemic control (admission glycated hemoglobin level: 8.20%). Hypertension, diagnosed over a decade ago, was also present, with peak blood pressure reaching 186/96 mmHg, and has been treated chronically with nifedipine controlled-release tablets.

The patient developed left-sided hemiplegia following a prior cerebral hemorrhage. There is no history of tobacco use or alcohol consumption. No significant family medical history was reported. One month before symptom onset, the skin on the left lateral malleolus became ulcerated due to repeated friction from a hemiplegic brace on the left limb. Initial self-management proved ineffective. Subsequently, the patient underwent regular debridement and dressing changes, along with oral antibiotic therapy, at another hospital. However, the wound continued to exhibit purulent discharge and progressive enlargement. The patient is experiencing anxiety due to the prolonged course of illness.

Upon admission, physical examination revealed a skin defect measuring approximately 4 cm × 3 cm on the left lateral malleolus, accompanied by a small amount of purulent exudate and surrounding erythema and swelling. A latent cavity approximately 2 cm in depth was identified, with exposure of the fibula (Fig. 1).

Wound culture yielded *Escherichia coli*. Magnetic resonance imaging (MRI) of the left ankle showed soft tissue swelling extending into the defect. Vibration sensation tested with a 128 Hz tuning fork was diminished, the 10 g monofilament test was positive, Achilles tendon reflex was reduced, and dorsalis pedis arterial pulsations were palpable bilaterally. Head computed tomography (CT) showed encephalomalacic foci in the right external capsule, basal ganglia–corona radiata region, and frontal lobe, along with small calcified nodules in the right frontal lobe.



Fig. 1 The patient's wound status upon hospital admission

Course of treatment

Upon admission, the patient was administered intravenous antibiotics to control wound infection. Concurrently, comprehensive blood glucose management was initiated with acarbose and metformin. During the first operation, soft tissue debridement and infection control were performed. The wound was repeatedly irrigated with normal saline and iodophor to remove surface contaminants. Necrotic, infected tissues and any foreign matter were thoroughly debrided using a curette until healthy bone tissue was exposed. Antibacterial bone cement (Fig. 2A) was then applied to fill the wound, forming a protective separation against secondary infection. The wound was subsequently covered with a sterile dressing and wrapped with a pressurized elastic bandage.

During the operation, each perforating branch of the peritoneal artery was explored, and intraoperative findings were consistent with preoperative ultrasonic findings. A fascial pedicle flap based on the peritoneal artery was designed according to routine procedures. The flap rotation point was located 7 cm above the lateral malleolus, and the flap size was slightly larger than the defect area (Fig. 2B). All perforator vessels from the gastrocnemius muscle to the flap were ligated and severed. The sural nerve and small saphenous vein were preserved and attached to the fascia (Fig. 2C), and the flap was carefully dissected under the fascia. A rubber membrane was discontinuously sutured beneath the flap to prevent detachment or curling (Fig. 3A), followed by in situ suturing of the flap (Fig. 3B). This marked the completion of the first phase of the operation. Postoperatively, the blood supply of the flap was closely monitored, with checks on its color, temperature, and capillary refill every 2 hours. Attention was also paid to potential swelling, hematoma, or compromised circulation around the flap's vascular pedicle. Excessive activity was restricted to avoid



Fig. 2 **A** Antibacterial bone cement used for wound filling. **B** Flap design: fasciocutaneous flap based on the peroneal artery, with the pivot point located 7 cm above the lateral malleolus and the flap dimensions slightly larger than those of the defect area. **C** Flap separation: preserved the sural nerve and small saphenous vein, and incorporated them into the fascial layer



Fig. 3 **A** A rubber membrane was discontinuously sutured beneath the flap to prevent detachment or curling. **B** Suturing the flap into position. **C** The wound 10 days after the first stage of the operation was completed

affecting blood flow, and the stability of the rubber membrane was regularly checked to ensure the flap remained intact.

Ten days after the first phase of the operation (Fig. 3C), the delayed flap was separated along the original surgical incision and trimmed. The defective wound was repaired using the delayed flap. After tight closure of the donor site, a full-thickness inguinal skin graft was

harvested, and the remaining wound was closed with pressure dressing (Fig. 4A). Anticoagulation, antiinflammatory, and antispasmodic therapies were continued for 7 days postoperatively. Strict aseptic care was maintained to prevent contamination or bacterial infection. Regular assessments of the wound condition, flap perfusion, and timely dressing changes were conducted to avoid poor blood flow due to excessive local pressure. Stitches were



Fig. 4 **A** Tight closure of the donor site, harvesting of full-thickness inguinal skin graft, and closure of the remaining wound with pressure dressing. **B** Stitches were removed 14 days postoperatively, revealing complete survival of the skin flap

removed 14 days postoperatively, revealing complete survival of the skin flap (Fig. 4B). Follow-up examination 3 months postoperatively demonstrated that the skin flap had a soft texture and exhibited good recovery in both sensation and function. The patient expressed high satisfaction with the treatment outcomes.

Discussion and conclusion

The treatment of DFUs is challenging, especially in the context of lower limb ischemia, neuropathy, and systemic abnormalities in blood glucose and lipid levels. Wound healing is often limited by poor local blood supply, and patients with diabetes typically exhibit compromised immune function. Consequently, traditional flap repair techniques face significant risks, including insufficient blood flow, postoperative thrombosis, and flap necrosis, which negatively impact the survival rate and healing outcomes of the flap [5, 6]. Baumeister *et al.* investigated distal pedicled sural fasciocutaneous flaps in 70 high-risk patients, showing that 12 cases (17%) experienced partial or terminal necrosis, while 13 cases (18.6%) exhibited complete necrosis, resulting in a total necrosis rate of 36%. Six cases ultimately required amputation. Demiri *et al.* compared the efficacy of distal pedicled sural nerve fascial flaps and shank perforator-based propeller flaps

for repairing foot and ankle wounds in patients with diabetes. Their findings indicated that complete flap survival rate in both groups was below 60%, with flap necrosis being a common complication [7].

In addition to diabetes, the present patient also suffered from hypertension and left-sided hemiplegia following cerebral hemorrhage, further increasing the complexity of the surgery and postoperative care. Given the deep penetration of the wound into the joint cavity and the poor local blood supply, traditional repair methods proved ineffective. The separation-delayed skin flap repair technique demonstrates unique advantages in such cases. This technique enhances the blood supply of the flap in stages, promoting angiogenesis and collateral circulation formation, thereby improving the survival rate of the flap [8, 9]. During the initial phase of the operation, by temporarily restricting part of the blood supply, the flap relies on alternative vascular sources, stimulating angiogenesis and collateral circulation development while enhancing its tolerance to ischemia, and 1–2 weeks after surgery, the blood supply undergoes remodeling, significantly improving the flap's survival rate and reducing the risks of necrosis and infection [8]. For high-risk patients, particularly those with dyskinesia and self-care difficulties, the separation-delayed flap technique offers

substantial benefits. Through the pre-adaptation process, the flap gradually adapts to the new blood supply pattern, markedly decreasing the incidence of postoperative complications. Roberts and DeSilva's study demonstrated a 94% success rate for delayed distal pedicle sural fasciocutaneous flaps, further validating the efficacy and potential of this technique [10].

In summary, the separation-delayed flap technique provides an effective solution for the complex wound repair of diabetic foot ulcers, especially in patients with severe vascular disease and additional comorbidities. By optimizing blood supply and promoting microcirculation, this technique significantly enhances the survival rate of the skin flap and exhibits broad clinical application potential.

Abbreviation

DFU Diabetic foot ulcer

Acknowledgements

The authors would like to acknowledge Dr. Rong Wang and Yunfu Zeng for support in imaging collection. This research was supported by the Academician Innovation Platform of Hainan Province, National Natural Science Foundation of China (82560446), Natural Science Foundation of Hainan Province (822MS174), (822RC692), and the Science and Technology Special Fund of Hainan Province (ZDYF2025SHFZ049)

Author contributions

Guan Aijun drafted the manuscript. Cheng Shaowen, Bian Yangyang, and Yao Jiangling performed the operation. Zeng Dongyang, Cheng Shaowen, and Wang Danxin provided critical guidance on the manuscript. Huang Hong, Jiang Meiting, and Cai Xin organized the materials. All authors read, reviewed, and approved the final version of the article.

Funding

No funding was received for this article.

Data availability

Datasets used during the current study period are available from the appropriate authors upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the principles of the Declaration of Helsinki. Written informed consent was obtained from the participant, and patient anonymity was strictly preserved.

IRB approval

This article was approved by the IRB of the First Affiliated Hospital Medical Center of Hainan Medical University.

Consent for publication

Written informed consent was obtained from the patient for the publication of this case report and any accompanying images. A copy of the written consent form is available for review by the Editor-in-Chief of this journal upon request.

Competing interests

The authors declare that they have no competing interests.

References

1. Sun H, Saeedi P, Karuranga S, et al. IDF Diabetes Atlas: Global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045. *Diabetes Res Clin Pract.* 2022;183: 109119.
2. van Netten JJ, Bus SA, Apelqvist J, et al. Definitions and criteria for diabetic foot disease. *Diabetes Metab Res Rev.* 2020;36(Suppl 1): e3268.
3. Armstrong DG, Boulton AJ, Bus SA. Diabetic foot ulcers and their recurrence. *N Engl J Med.* 2017;376(24):2367–75.
4. Ziegler T, Abdel Rahman F, Jurisch V, et al. Atherosclerosis and the capillary network; pathophysiology and potential therapeutic strategies. *Cells.* 2019. <https://doi.org/10.3390/cells9010050>.
5. Brem H, Tomic-Canic M. Cellular and molecular basis of wound healing in diabetes. *J Clin Invest.* 2007;117(5):1219–22.
6. Zhu Y, Wang Y, Jia Y, et al. Roxadustat promotes angiogenesis through HIF-1 α /VEGF/VEGFR2 signaling and accelerates cutaneous wound healing in diabetic rats. *Wound Repair Regen.* 2019;27(4):324–34.
7. Demiri E, Tsimponis A, Pavlidis L, et al. Reverse neurocutaneous vs propeller perforator flaps in diabetic foot reconstruction. *Injury.* 2020;51(Suppl 4):S16–21.
8. Saint-Cyr M, Wong C, Schaverien M, et al. The perforasome theory: vascular anatomy and clinical implications. *Plast Reconstr Surg.* 2009;124(5):1529–44.
9. Taylor GI, Corlett RJ, Ashton MW. The functional angiosome: clinical implications of the anatomical concept. *Plast Reconstr Surg.* 2017;140(4):721–33.
10. Roberts HJ, DeSilva GL. Can sural fasciocutaneous flaps be effective in patients older than 65? *Clin Orthop Relat Res.* 2020;478(4):734–8.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.